

WASP-43b

R-2667

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_130511 · created 2026-08-30T00:57:47+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456423.68911 +/- 0.00058 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.02
Transit depth (Rp/Rs)^2	3.91 +/- 0.78 [%]
Orbital Inclination (inc)	84.15 +/- 1.96
Airmass coefficient 1 (a1)	0.999 +/- 0.042
Airmass coefficient 2 (a2)	-0.001 +/- 0.016
Scatter in the residuals of the lightcurve fit is	4.2 %
Best Comparison Star	#2 - [592, 224]
Optimal Aperture	2.99
Optimal Annulus	13.27
Transit Duration (day)	0.0578 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.198 ± 0.02 vs 0.1594 (deviation 1.33σ)
Duration fit vs published	0.0578 d vs 0.0512 d (deviation 1.0σ)
Mid-time O–C	-1.6 min
Depth SNR	6.8
Residual scatter	4.2 %

Light curve

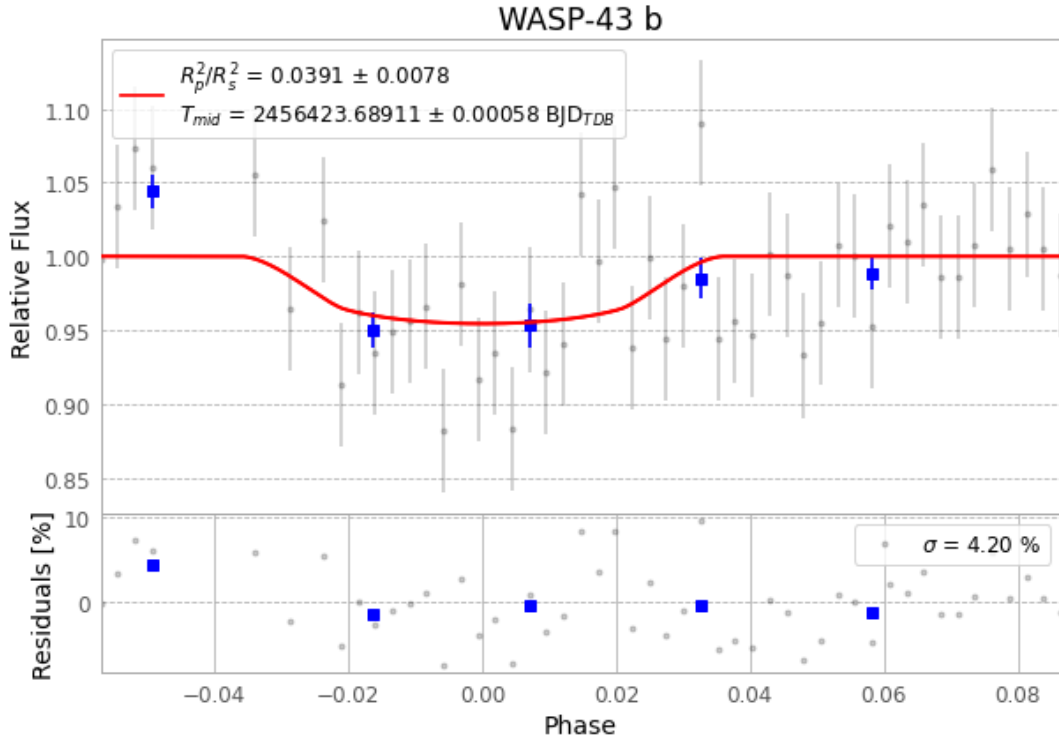


Figure 1 — FinalLightCurve_WASP-43 b_2013-05-10.png (SHA-256 156ca40dc279f5d2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [316, 314], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 212cfc8d34b27bbe...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

57 FITS frames (37 MB), WASP-43130511032112.FITS → WASP-43130511060912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-130511.log (e58e3a2d27ae...), FinalLightCurve_WASP-43 b_2013-05-10.png (156ca40dc279...), FinalLightCurve_WASP-43 b_2013-05-10.csv (ae7c8331b217...)

Compute resources

Wall time 349.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 00:57Z.